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An iron-containing POM-based hybrid compound as a heterogeneous catalyst for one-step hydroxylation of benzene to phenol†

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It is a major challenge to perform one-pot hydroxylation of benzene to phenol under mild conditions, which replaces the environmentally harmful cumene method. Thus, finding highly efficient heterogeneous catalysts that can be recycled is extremely significant. Herein, a (POM)-based hybrid compound $\{[\text{Fe}^{\text{II}}(\text{pyim})_2(\text{C}_2\text{H}_5\text{O})][\text{Fe}^{\text{II}}(\text{pyim})_2(\text{H}_2\text{O})][\text{PMo}_2^{\text{V}}\text{Mo}_9^{\text{V}}\text{V}_3^{\text{V}}\text{O}_{42}]\cdot\text{H}_2\text{O}$ (pyim = 2-(2-pyridyl)benzimidazole) (**Fe₂-PMo₁₁V₃**) was successfully prepared by hydrothermal synthesis using typical Keggin POMs, iron ions and pyim ligands. Single-crystal diffraction shows that the Fe-pyim unit in **Fe₂-PMo₁₁V₃** forms a stable double-supported skeleton by Fe–O bonding to the polyacid anion. Remarkably, due to the introduction of vanadium, **Fe₂-PMo₁₁V₃** forms a divanadium-capped conformation. Benzene oxidation experiments indicated that **Fe₂-PMo₁₁V₃** can catalyze the benzene hydroxylation reaction to phenol in a mixed solution of acetonitrile and acetic acid containing H₂O₂ at 60 °C, affording a phenol yield of about 16.2% and a selectivity of about 94%.

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Introduction

As one of the indispensable organic chemical intermediates, phenol is widely used as an important raw material for the production of certain resins, fungicides, preservatives and drugs.^{1,2} Back in the twentieth century, Hock reported the production of phenol by a three-step cumene process,³ and about 90% of the phenol currently produced in the industry is obtained by the three-step cumene method.^{4–8} However, this conventional method requires high energy consumption and can produce toxic intermediates and equal amounts of benzoquinone as a by-product.⁹ Thus, it is necessary to develop an efficient and environmentally friendly method to produce phenol by one-pot direct hydroxylation of benzene. The catalyst for one-pot benzene hydroxylation to phenol is inextricably related to the oxidant used, with the commonly used oxidants mainly being N₂O, H₂O₂, O₂, etc.^{10–13} Due to the high activity exhibited by H₂O₂,^{14,15} efforts have been made to develop cata-

lysts for the catalytic hydroxylation of benzene to phenol using H₂O₂ as an oxidant. Benzene hydroxylation is a reaction that activates the C–H bond of the benzene ring; this reaction process is difficult to be successfully complete under mild liquid phase conditions because the benzene ring has conjugated large π -bonds with high chemical stability. Hence, the search for highly active catalysts for the direct hydroxylation of benzene to phenol is a major challenge for academia as well as the industry.^{16–19}

Polyoxometalates (POMs), as a class of inorganic metal oxide clusters with a rich structure and excellent redox properties, have attracted broad attention in the fields of catalysis, optics, magnetism, etc.^{20–32} Among numerous POM-based materials, molybdovanadophosphoric acid (PMoV) with the Keggin structure has attracted the attention of a wide range of researchers. Since this form of polyacid has abundant and variable metal valence states, it can activate molecular oxidation to obtain reactive oxygen species and easily selectively oxidize reactive organic substrates without the addition of reducing agents.^{33,34} Until now, Mo–V miscible heteropolyacids have been applied in the oxidation of benzene to phenol by a catalyst and have shown high activity and selectivity. For example, Ronny Neumann used H₅PMo₁₀V₂O₄₀ as the catalyst and 50% H₂SO₄ solution as a solvent, and the highly selective hydroxylation of benzene occurred in the presence of O₂.³⁵ Wang *et al.* utilized a mixture of g-C₃N₄ and PMo₁₀V₂ as a catalyst for the reaction of benzene with oxygen and obtained a high yield of phenol (10.7%).³³ Just recently, Wei *et al.* reported a photo-

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catalytic system constructed with alkoxyhexavanadate anions and quinoline ions, and a high phenol yield (50.1%) and selectivity (99%) were achieved in the H₂SO₄-containing reaction system.³⁶ However, like other homogeneous catalysts, this catalyst is difficult to separate from the reaction medium. Hence, it is vital to develop reusable and highly active catalysts.

Molybdovanadophosphoric acid (PMoV) is usually soluble in different solvents and unfavorable for catalyst recirculation, thus, we introduced iron and an organic ligand to construct an Fe-capped Keggin-type POM hybrid. Moreover, iron-based catalysts have been confirmed to be effective catalysts for the direct hydroxylation of benzene to phenol.^{37–39} And the organic nitrogen heterocyclic ligands make conjugated π electrons more active, thereby activating the C–H bond of the benzene ring.^{40,41} Based on the above analysis, a POM-based hybrid compound $\{[\text{Fe}^{\text{II}}(\text{pyim})_2(\text{C}_2\text{H}_5\text{O})][\text{Fe}^{\text{II}}(\text{pyim})_2(\text{H}_2\text{O})][\text{PMo}_2^{\text{V}}\text{Mo}_9^{\text{VI}}\text{V}_3^{\text{IV}}\text{O}_{42}]\} \cdot \text{H}_2\text{O}$ (**Fe₂-PMo₁₁V₃**) was synthesized *via* hydrothermal synthesis selecting pyim ligands and iron ions to modify the POM anion. **Fe₂-PMo₁₁V₃** exhibits an isolated structure, in which two Fe-pyim fragments are attached to the $[\text{PMo}_{11}\text{V}_3\text{O}_{42}]^{3-}$ anion with a bi-capping structure by the Fe–O bond. And **Fe₂-PMo₁₁V₃** was characterized by single crystal X-ray diffraction, FT-IR, and XPS. We found that benzene can be efficiently hydroxylated to phenol in the presence of H₂O₂ using **Fe₂-PMo₁₁V₃** as a catalyst and acetonitrile as a solvent. In addition, the catalytic performance is further improved when a small amount of acetic acid is added as a co-solvent. The results showed that the yield of phenol reached 16.2% and the selectivity was 94% in 6 h at 60 °C. Furthermore, the catalytic activity of **Fe₂-PMo₁₁V₃** hardly decreased after five cycles.

Experimental section

Materials and methods

H₆PMo₉V₃O₄₀·32.5H₂O was prepared according to the method provided in ref. 42.⁴² All other chemicals were commercially available and used without further purification. 5,5-Dimethyl-1-pyridine *N*-oxide (DMPO), isopropanol (IPA) and butylhydroxytoluene (BHT) were purchased from Aladdin Industries. 2-(2-Pyridyl)benzimidazole (pyim) and FeCl₃ were procured from Shanghai McLean Biochemical Co. Acetic acid, 30 wt% H₂O₂, acetonitrile and benzene were purchased from Shanghai Lingfeng Chemical Reagent Co.

Synthesis of $\{[\text{Fe}^{\text{II}}(\text{pyim})_2(\text{C}_2\text{H}_5\text{O})][\text{Fe}^{\text{II}}(\text{pyim})_2(\text{H}_2\text{O})][\text{PMo}_2^{\text{V}}\text{Mo}_9^{\text{VI}}\text{V}_3^{\text{IV}}\text{O}_{42}]\} \cdot \text{H}_2\text{O}$

A mixture of H₆PMo₉V₃O₄₀ (0.10 g, 0.06 mmol), 2-(2-pyridyl)benzimidazole (0.02 g, 0.10 mmol), FeCl₃·6H₂O (0.10 g, 0.37 mmol), H₂O (8 mL) and ethanol (2 mL) was stirred for 60 min at room temperature, and then the pH value was adjusted at 2.0 by adding HCl (1 M). Whereafter, the mixture was placed into a 25 mL Teflon-lined autoclave, which was heated at 170 °C for 72 h. It was allowed to cool to room temperature

and washed several times with distilled water and ethanol, and finally black block crystals (yield: 36% based on pyim) were obtained (Fig. S1†). Elemental analysis calcd (%): C 20.85, H 1.70, N 5.84; found (%): C 21.05, H 1.64, N 5.78.

Benzene hydroxylation to phenol

The process of hydroxylation to phenol was carried out in a 25 mL flask and heated in an oil bath. Firstly, the temperature of the oil bath was set to a constant value at which the entire catalytic reaction proceeded. The reaction mixture containing 10 mmol of benzene, 30 mmol of 30% H₂O₂, 10 mg of catalyst, 5 mL of acetonitrile and 1 mL of acetic acid was stirred constantly at the set temperature for 6 h. The cooled mixture was analyzed by gas chromatography (GC). According to the above analysis, the main products were phenol and benzoquinone.

The stability of **Fe₂-PMo₁₁V₃** was evaluated in the recycling experiment. After the reaction of catalyst recycling, it was filtered and repeatedly washed with distilled water and ethanol, dried at 50 °C, and then employed in the next cycle.

Results and discussion

Synthesis

Hydrothermal synthesis is known to be an effective way to synthesize POM-based derivatives, and the reaction time, temperature, pH value and the amount of reactants are the key factors that affect crystal growth. In the synthesis process, we found that the temperature and pH value have obvious effects on crystal formation. The optimum reaction temperature for crystal formation is 170 °C, but a small number of crystals appear at 180 °C. The optimum pH value of the reaction is 2.0, and crystals cannot be obtained when the pH is less than 1.8 or greater than 2.5. Moreover, **Fe₂-PMo₁₁V₃** has good repeatability.

Crystal structure

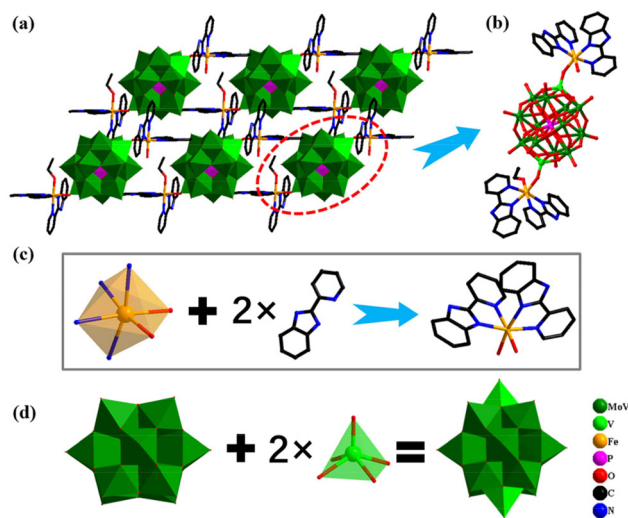
X-ray analysis of the single crystal shows that **Fe₂-PMo₁₁V₃** crystallizes in the monoclinic space group *P*1̄ (Table 1). The molecular structure of **Fe₂-PMo₁₁V₃** consists of a $[\text{PMo}_2^{\text{V}}\text{Mo}_9^{\text{VI}}\text{V}_3^{\text{IV}}\text{O}_{42}]^{3-}$ anion, a $[\text{Fe}^{\text{II}}(\text{pyim})_2(\text{C}_2\text{H}_5\text{O})]^+$ cation fragment, a $[\text{Fe}^{\text{II}}(\text{pyim})_2(\text{H}_2\text{O})]^{2+}$ cation fragment and a lattice water molecule (Fig. 1b). The $[\text{Fe}^{\text{II}}(\text{pyim})_2(\text{CH}_3\text{O})]^+$ fragment consists of an iron ion, two pyim ligands and an ethanol molecule, while the $[\text{Fe}^{\text{II}}(\text{pyim})_2(\text{H}_2\text{O})]^{2+}$ fragment is the same as the above except that the ethanol molecule is replaced by the coordinated water molecule. The two fragments are both connected to capped-Keggin type $[\text{PMo}_2^{\text{V}}\text{Mo}_9^{\text{VI}}\text{V}_3^{\text{IV}}\text{O}_{42}]^{3-}$ through Fe–O bonds, forming an interesting double support structure. The two iron atoms in **Fe₂-PMo₁₁V₃** have the same coordination mode: the two iron ions reside in a distorted octahedron, where the hexa-coordinated Fe atoms are ligated by two O atoms originating from the ethanol/water molecule and μ_2 -O and four N atoms from the two pyim ligands (Fig. 1c).

Notably, the principal building unit presents an intriguing bicapped configuration due to the introduction of vanadium

Table 1 Crystallographic data for the compound **Fe₂-PMo₁₁V₃**

Compound	Fe₂-PMo₁₁V₃
Formula	C ₅₀ H ₄₅ Fe ₂ Mo ₁₁ N ₁₂ O ₄₅ PV ₃
Formula weight	2884.81
T (K)	296(2)
Crystal system	Triclinic
Space group	<i>P</i> $\bar{1}$
<i>a</i> (Å)	12.407(3)
<i>b</i> (Å)	13.254(4)
<i>c</i> (Å)	13.666(4)
α (°)	76.214(3)
β (°)	83.441(3)
γ (°)	65.960(3)
<i>V</i> (Å ³)	1992.8(10)
<i>Z</i>	1
<i>D_c</i> (mg cm ⁻³)	2.397
μ (mm ⁻¹)	2.470
<i>F</i> (000)	1365.1
θ range (°)	0.903–25.008
Crystal size (mm ³)	0.14 × 0.13 × 0.12
Limiting indices	–13 ≤ <i>h</i> ≤ 15 –16 ≤ <i>k</i> ≤ 16 –16 ≤ <i>l</i> ≤ 16
Reflections collected	13 908
Independent reflection/unique	7249
<i>R</i> (int)	0.0618
Data/restraints/parameters	7249/18/598
GOF	1.112
<i>R</i> ₁ ^a , <i>wR</i> ₂ ^b [<i>I</i> > 2σ(<i>I</i>)]	<i>R</i> ₁ = 0.0514 <i>wR</i> ₂ = 0.1443
<i>R</i> ₁ , <i>wR</i> ₂ (all data)	<i>R</i> ₁ = 0.0706 <i>wR</i> ₂ = 0.1598

$$^a R_1 = \sum ||F_o| - |F_c|| / \sum |F_o|, \quad ^b wR_2 = \sum [w(F_o^2 - F_c^2)^2] / \sum [w(F_o^2)^2]^{1/2}.$$

**Fig. 1** (a) Stick and polyhedral diagram of the two-dimensional supramolecular layer; (b) ball-and-stick diagram of **Fe₂-PMo₁₁V₃**; (c) ball-and-stick diagram of the [Fe^{II}(pyim)₂(CH₃O)]⁺ fragment; (d) the divanadium-capped configuration of **Fe₂-PMo₁₁V₃**.

in H₆PMo₉V₃O₄₀·32.5H₂O, in which two {VO₅} tetragonal pyramids are connected to the Keggin-type [PMo₁₁VO₄₀]⁷⁻ anion by the edge-sharing mode (Fig. 1d). Moreover, all Mo and V in the [PMo₁₁VO₄₀]⁷⁻ anion are disordered. The previously reported compound [Fe(DAPSC)(H₂O)₂][HPMo₂Mo₁₀^{VI}O₄₀]₂·5H₂O is

different from **Fe₂-PMo₁₁V₃**, in which the polyacidic anion connects the two [Fe^{II}(pyim)₂(CH₃O)]⁺ cationic fragments *via* the Fe–O bond.⁴³

The adjacent basic units form a 2D supramolecular structure through weak interactions of hydrogen bonds (Fig. S3†). The adjacent 2D supramolecular layers are further connected by hydrogen bonds to form a 3D supramolecular structure (Fig. S4†). For **Fe₂-PMo₁₁V₃**, the Fe–O and Fe–N lengths are in the ranges 2.113(6)–2.157(7) Å and 2.119(7)–2.228(7) Å (Table S1†); besides, the bond angles O–Fe–O, N–Fe–O, and N–Fe–N are 87.7(3)°, 91.3(3)–172.4(3)°, and 76.2(3)–166.0(3)° (Table S2†). The bond lengths and angles of **Fe₂-PMo₁₁V₃** are within a reasonable range for other Fe-containing complexes.^{44,45} For **Fe₂-PMo₁₁V₃**, bond valence sum (BVS) calculations (Table S4†) show that the Fe ions are in the +2 valence state, which is consistent with the X-ray photoelectron spectroscopy (XPS) data (Fig. S8b†). Furthermore, Mo(1)–Mo(6) are all occupied by Mo and V. For the Mo and V center oxidation states, bond valence sum (BVS) calculations reveal that two V(7) atoms at both ends and disordered V atoms are assigned to be V^{IV} centers, and Mo(1, 2, 3, 4, and 6) and Mo(5) are assigned the values Mo^{VI} and Mo^V, respectively (Table S4†),⁴⁶ in consistence with the XPS data (Fig. S8†). BVS calculations of the Mo coordination spheres provides values of 6.165 for Mo(1), 5.891 for Mo(2), 5.920 for Mo(3), 6.039 for Mo(4), 5.381 for Mo(5) and 5.945 for Mo(6), and the average oxidation state is 5.89 (the expected value is 5.88), indicating that the 4d electrons are delocalized in the capped-Keggin unit.

The XRD and IR analysis

We irradiated the sample using an X-ray diffractometer at room temperature, as illustrated in Fig. S5,† and the X-ray diffraction pattern showed that the main peaks of **Fe₂-PMo₁₁V₃** agreed well with the theoretically simulated peaks, proving that **Fe₂-PMo₁₁V₃** was a pure phase.

The IR spectrum of **Fe₂-PMo₁₁V₃** in the range of 4000–400 cm⁻¹ is shown in Fig. S6.† A broad peak at 3608 cm⁻¹ shows the presence of water, and the peaks at 1059 cm⁻¹ and 961 cm⁻¹ correspond to the characteristic absorption peaks of P–O and Mo=O. The absorption peaks at 876 cm⁻¹ and 788 cm⁻¹ correspond to the asymmetric Mo–O–Mo vibrations, which are all attributed to the characteristic peaks of PMo₁₁V₃O₄₂ anions.⁴⁷ In addition, the peak at 2975 cm⁻¹ is attributed to the dissymmetric and symmetric stretching vibration of C–H on the organic ligand pyim, the peak near 1604 cm⁻¹ is attributed to the C=N stretching vibration, the peak at 1438 cm⁻¹ is due to the C=C stretching vibration of the benzene ring on the ligand, and the peak at 1301 cm⁻¹ is attributed to the C–N stretching vibration, and the above characteristic peaks are attributed to the characteristic peaks of the organic ligand pyim.⁴⁸

Catalytic activity in benzene hydroxylation

The catalytic performance of **Fe₂-PMo₁₁V₃** was investigated with H₂O₂ as the oxidant. As presented in Table 2 for entries 1 and 7, almost no phenol was detected in the absence of an

Table 2 Effect of H₂O₂/benzene at different molar ratios on the oxidation of benzene to phenol^a

Entry	Molar ratio (H ₂ O ₂ /benzene)	Phenol yield (%)	Sel. (%)	H ₂ O ₂ utilization (%)
1	(0/1)	<0.1	—	—
2	(1/1)	3.7	92.2	4.5
3	(2/1)	7.5	93.1	4.4
4	(3/1)	16.2	94.0	6.3
5	(4/1)	13.8	93.5	4.0
6	(5/1)	7.3	91.0	1.7
7 ^b	(3/1)	<0.1	—	—

^a Reaction conditions: solvent of acetonitrile and acetic acid (6 mL, volume ratio 5 : 1), catalyst (10 mg), temperature (60 °C), reaction time (6 h). ^b Without catalyst.

oxidant or a catalyst, revealing that the catalyst and oxidant play an indispensable role in the reaction process. Then we explored the optimal reaction conditions by optimizing the catalyst dosage, oxidant dosage, reaction time and temperature to obtain a higher conversion of benzene and selectivity of phenol. First, the effect of catalyst dosage on the benzene hydroxylation reaction was investigated. The phenol yield increased with the increasing catalyst amount until an optimum catalyst amount was reached, as the higher catalyst dosage resulted in more catalytically active sites (Fig. 2a). The highest phenol yield of 11.2% and selectivity of 92.6% were achieved when the catalyst dosage reached 10 mg. With further increase in the catalyst dosage, the phenol yield decreased significantly and the selectivity decreased slightly; the catalyst dosage thus resulted in phenol oxidation. Hence, we subsequently chose the optimal catalyst dosage of 10 mg.

Also, temperature is also one of the important factors affecting the catalytic activity. The phenol yield as well as the selectivity increased with the increase in temperature, and the

phenol yield reached the highest value of 12.6% when the temperature reached 60 °C (Fig. 2b). When the temperature increased further the phenol yield decreased instead, and the selectivity reached the highest value of 93% at 60–70 °C. When the temperature exceeded 70 °C, the phenol yield decreased significantly, which may be due to the autolysis of H₂O₂ at higher temperatures, thus causing significant losses. Therefore, we set the temperature of the subsequent catalytic experiment to 60 °C.

Furthermore, the reaction time had an effect on the phenol yield, and the effect of the reaction time on benzene hydroxylation was examined in the range of 4–8 h. The yield and selectivity of phenol increased with the extension of the reaction time, and the phenol yield reached a maximum of 16.2% and the phenol selectivity reached 94% when the reaction time reached 6 h (Fig. 2c). Then, the phenol yield and selectivity decreased with the increase of the reaction time, which was because the generated phenol was further oxidized. It could be speculated that when the hydroxylation time was too long, there was a greater possibility that the large amount of radicals generated would oxidize phenol rather than benzene. Therefore, considering the cost and effect of the subsequent catalytic experiments, we chose a reaction time of 6 h.

In addition to the reaction time and temperature, the oxidant is an integral part of the catalytic process. Therefore, the effect of different oxidant dosages on the hydroxylation of benzene was studied and the effect of different molar ratios of H₂O₂ to benzene on the benzene hydroxylation activity was compared under the above conditions. As presented in Table 2 entries 1–4, as the amount of oxidant increased, the phenol yield increased and there was a slight increase in the selectivity. The highest phenol yield was 16.2% when the molar ratio of H₂O₂ to benzene reached 3 : 1. It was found that the yield and selectivity of phenol decreased if the amount of oxidant dosage continued to increase (entries 4–6), which may be due to the presence of a large amount of H₂O₂ in the reaction solution and a small amount of benzene in the presence of sufficient H₂O₂. Once the benzene was completely converted into phenol, the phenol continued to react with the oxidant to form benzoquinone. So, we chose a H₂O₂ to benzene molar ratio of 3 : 1 as the optimal dosage.

Finally, we investigated the effect of the acetonitrile/acetic acid volume ratio on benzene oxidation. As shown in Table 3, the solvent did not contain acetic acid, and the phenol yield was still 7.5%, however, the selectivity was only 78% (entry 1), which indicated that hydroxylation reaction of benzene was not dependent on acetic acid, but only provided an acidic medium for the reaction. When acetic acid was added as a binary solvent, the volatilization of benzene as well as the oxidation of phenol were inhibited. The phenol yield and selectivity increased from 7.5% and 78% to 16.2% and 94%, respectively (entry 2). As indicated in entries 2–5, as the amount of acetic acid increased, the yield of phenol decreased continuously. Thus, benzene oxidation and yield of phenol were significantly inhibited when acetic acid was in excess. Consequently, the highest yield of phenol was obtained when

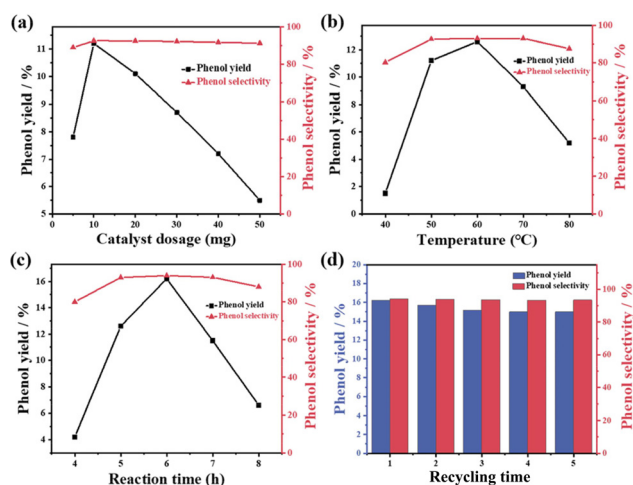


Fig. 2 Effect of (a) catalyst dosage, (b) reaction temperature and (c) reaction time on the benzene hydroxylation reaction. (d) Reusability of Fe₂-PMo₁₁V₃. Reaction conditions: benzene (10 mmol), acetic acid (1 mL), acetonitrile (5 mL), H₂O₂ (30 mmol), 60 °C, 6 h.

the ratio of water to acetic acid was 5 : 1. Ethanol, methanol and dichloromethane can also be used as reaction solvents for the hydroxylation reaction of benzene to phenol, so we conducted comparative experiments for solvent screening. Obviously, the acetonitrile environment could afford a more gratifying yield under the same conditions (Fig. S11†). We explored the effect of catalyst precursors such as FeCl_2 , pyim, and PMo_9V_3 as catalysts on benzene oxidation performance, respectively. As shown in entries 6–8 in Table 3, it was found that the ligand pyim itself is not catalytically active and pure PMo_9V_3 and FeCl_2 were difficult to recover as homogeneous

catalysts although they could improve the phenol yield to some extent.

Apart from the excellent phenol yield, the reusability and stability of the catalyst were important factors for heterogeneous catalysts. Hence, the stability of $\text{Fe}_2\text{-PMo}_{11}\text{V}_3$ as a catalyst after five cycles was explored under the above optimal reaction conditions. At the end of each reaction, the catalyst was easy to separate from the system and cleaned with acetonitrile. The dried catalyst could be directly reused in subsequent reactions. As shown in Fig. 2d, there was almost no distinct decrease in the yield and selectivity of phenol after five cycles. In addition, the XRD patterns and IR spectra of $\text{Fe}_2\text{-PMo}_{11}\text{V}_3$ before and after the reaction indicated a retention in the crystallinity (Fig. 3). The high stability of $\text{Fe}_2\text{-PMo}_{11}\text{V}_3$ as a heterogeneous catalyst for the benzene hydroxylation reaction was demonstrated.

Proposed mechanism

As demonstrated in Fig. 4a, the benzene oxidation reaction was completely inhibited without the addition of H_2O_2 , indicating the importance of H_2O_2 in our reaction system. To investigate the active substances in the benzene oxidation process, isopropanol (IPA, $\cdot\text{OH}$ scavenger) and butylhydroxytoluene (BHT, $\cdot\text{O}_2$ scavenger) were added in equal amounts to the initial reaction solution, and free radical trapping experiments were carried out with other reaction conditions remaining unchanged, and the yields of phenol were reduced to 2.3% and 14.1%, respectively. The oxidation reaction was inhibited considerably by the addition of IPA, and was only slightly inactivated in the presence of BHT. This indicated that the main active substance in the reaction system might be $\cdot\text{OH}$.

In order to confirm the existence of the hydroxyl radical, PL spectroscopy was further performed. Currently, coumarin has been proved to be used as a probe molecule to detect free-state $\cdot\text{OH}$ in the reaction solution.⁴⁹ Coumarin itself does not have fluorescence properties, but the reaction between coumarin and $\cdot\text{OH}$ can generate a variety of OH-substituted coumarins, in which umbelliferone can emit strong fluorescence. In Fig. 4b, the PL spectra before and after the reaction with the

Table 3 Benzene oxidation to phenol in various volumes of acetic acid^a

Entry	Acetonitrile/acetic acid (mL/mL)	Phenol yield (%)	Sel. (%)
1	(5/0)	7.5	78.0
2	(5/1)	16.2	94.0
3	(5/2)	15.0	94.0
4	(5/3)	13.5	93.8
5	(5/4)	10.0	94.2
6 ^b	(5/1)	13.0	96.0
7 ^c	(5/1)	12.7	96.5
8 ^d	(5/1)	<0.1	—

^a Reaction conditions: benzene (10 mmol), 30% H_2O_2 (30 mmol), catalyst (10 mg), temperature (60 °C), reaction time (6 h). ^b $\text{H}_6\text{PMo}_9\text{V}_3\text{O}_{40}$ as the catalyst. ^c FeCl_2 as the catalyst. ^d Pyim as the catalyst.

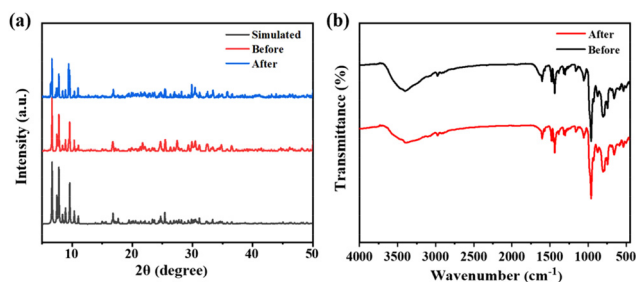


Fig. 3 (a) XRD patterns and (b) IR spectra of $\text{Fe}_2\text{-PMo}_{11}\text{V}_3$ before and after benzene oxidation reaction.

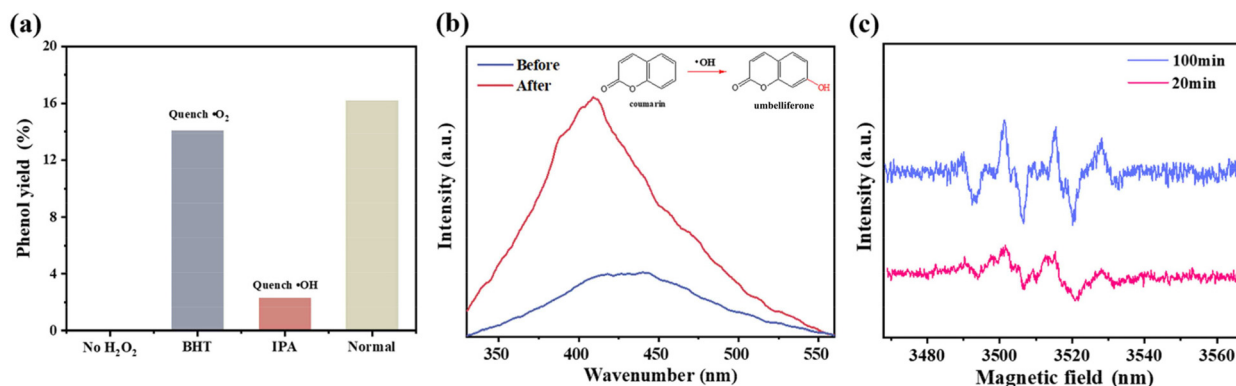
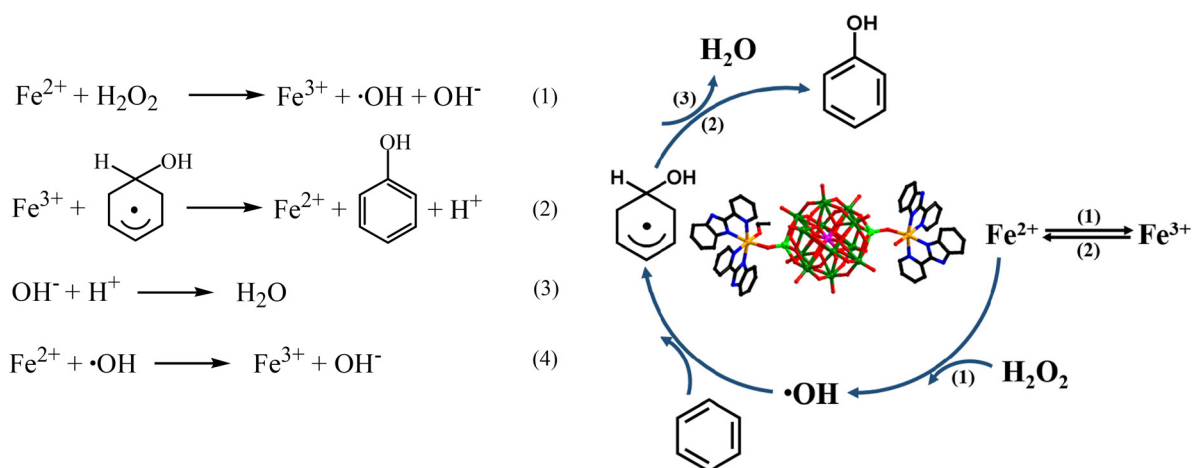


Fig. 4 (a) Benzene oxidation experiments with the addition of different radical scavengers; (b) PL spectra of coumarin in the reaction as a result of $\cdot\text{OH}$ trapping; and (c) EPR spectra of $\text{Fe}_2\text{-PMo}_{11}\text{V}_3$ in the presence of DMPO.



Scheme 1 Proposed mechanism for the benzene hydroxylation over $\text{Fe}_2\text{-PMo}_{11}\text{V}_3$.

Table 4 Comparison of the catalytic activity with some heterogeneous POM-based catalysts

Catalyst	Time/h	$T/^\circ\text{C}$	Phenol yield/%	Selectivity/%	Ref.
$\text{PMo}_{10}\text{V}_2/\text{pg-C}_3\text{N}_4$	8	60	25.7	99.7	55
$\text{H}_5\text{PMo}_{10}\text{V}_2\text{O}_{40}/\text{UiO-66-NH}_2$	4	60	14.08	98.8	56
$\text{HPMoV}[\text{ethanol}]/\text{MCM-41-NH}_2$	9	70	20.4	97.4	57
$[(\text{CH}_3)_4\text{N}]_4\text{PMo}_{11}\text{VO}_{40}$	2	60	12.4	85.7	58
$\text{PMoV}_2\text{-IL-Al-MCM-41}$	9	60	14.8	100	59
$[(\text{CH}_3)_4\text{N}]_6[\text{PMo}_9\text{V}_3\text{O}_{40}][\text{Cu}_2(\text{C}_6\text{H}_5\text{O}_6)_{4/3}]_6$	1.3	80	9.93	99.0	60
$\{\text{Fe}^{\text{II}}(\text{pyim})_2(\text{C}_2\text{H}_5\text{O})\}[\text{Fe}^{\text{III}}(\text{pyim})_2(\text{H}_2\text{O})][\text{PMo}_2\text{Mo}_9\text{V}_3^{\text{VI}}\text{O}_{42}]\cdot\text{H}_2\text{O}$	6	60	16.2	94.0	This work

addition of coumarin are presented. Obviously, the fluorescence intensity increased significantly because of the formation of umbelliferone resulting from the adsorption of $\cdot\text{OH}$. That is, $\cdot\text{OH}$ is an indispensable intermediate in the reaction process.

Furthermore, spin trapping is also a method used to detect $\cdot\text{OH}$. We employed 5,5-dimethyl-1-pyrroline *N*-oxide (DMPO) as a radical trapping agent, which was added to the reaction solution after 20 min and 100 min of reaction, respectively. Subsequently, the potentials of the DMPO-radical adducts were measured using EPR spectroscopy. As displayed in Fig. 4c, four peaks with a ratio of 1:2:2:1 were found to emerge in both cases, and the signal of the DMPO- $\cdot\text{OH}$ adduct was significantly stronger at 100 min than that at 20 min. This again confirmed that phenyl hydroxylation on $\text{Fe}_2\text{-PMo}_{11}\text{V}_3$ follows an $\cdot\text{OH}$ radical-mediated pathway.

Based on the above results and the mechanism studies in the literature,^{50–54} a possible reaction pathway for the hydroxylation of benzene with POM-based catalysts was proposed (Scheme 1). Initially, Fe^{2+} in the compound was oxidized to Fe^{3+} by H_2O_2 and $\cdot\text{OH}$ was produced (eqn (1)). The generated $\cdot\text{OH}$ attacked the benzene ring, causing the C=H bond on the benzene ring to be activated and forming hydroxycyclohexadienyl radicals. Ultimately, hydroxycyclohexadienyl radicals reacted with Fe^{3+} to produce phenol and H_2O (eqn (2) and (3)). Besides, Fe^{3+} was reduced to Fe^{2+} , which regenerated the cata-

lyst, thereby completing the entire catalytic cycle. Although an appropriate increase in the concentration of Fe^{2+} could increase the rate of benzene oxidation, excessive Fe^{2+} would also scavenge $\cdot\text{OH}$ (eqn (4)),⁵³ which was not conducive to the oxidation of benzene.

A comparison of the catalytic activity of $\text{Fe}_2\text{-PMo}_{11}\text{V}_3$ with several previously reported POM-based heterogeneous catalysts in the presence of H_2O_2 is shown in Table 4.^{55–60} The phenol yield of $\text{Fe}_2\text{-PMo}_{11}\text{V}_3$ (16.2%) was higher than those of most of the POM-based catalysts presented in Table 4.

Conclusions

In summary, a capped-Keggin POM-based organic-inorganic hybrid compound was synthesized by the hydrothermal method, which was composed of the POM anion and the Fe-pyrim fragment, presenting a divanadium-capped structure. Specifically, a phenol yield of 16.2% and a selectivity of 94% were achieved within 6 h at 60 $^\circ\text{C}$ with H_2O_2 as the oxidant, and good activity was maintained after five cycles. In this study, $\text{Fe}_2\text{-PMo}_{11}\text{V}_3$ was considered as a benzene oxidation catalyst with the advantages of relatively easy synthesis, high stability and easy recyclability. We anticipate that this work can provide an effective way for the design and synthesis of POMs for realizing benzene oxidation to phenol.

Conflicts of interest

The authors declare no conflict of interest.

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